

Software Requirements Specification (SRS)

Project: PixiPlay - Mini Games Website

1. Introduction

PixiPlay is a web-based mini games platform where users can register, log in, play three classic games (2048, Sudoku, and Snake), track their scores, compete on a leaderboard, and earn bonus points through daily trivia challenges. The system integrates DiceBear Avatar API to generate unique user avatars and the Open Trivia Database API to provide quiz-based rewards.

2. Objectives

- Provide an engaging gaming platform
- Track scores and game history
- Display leaderboards
- Generate unique avatars automatically
- Reward users through trivia challenges

3. Users

Guest, Registered User, Administrator

4. Functional Requirements

- User registration/login
- Avatar generation using DiceBear API
- Play 2048, Sudoku, and Snake
- Save scores and game history
- Leaderboard
- Daily trivia challenge using Open Trivia DB
- User profile management

5. Non-Functional Requirements

Responsive UI, secure authentication, scalable REST APIs, maintainable architecture, fast game loading, reliable PostgreSQL storage.

6. Technology Stack

Frontend: React.js
Backend: Spring Boot
Database: PostgreSQL
ORM: Spring Data JPA
Security: Spring Security + JWT (optional/simple auth)
IDE: IntelliJ IDEA / VS Code

7. Database Tables

users, scores, game_history, leaderboard

8. External APIs

- DiceBear Avatar API – Generates avatars based on username.
- Open Trivia Database API – Provides trivia questions for bonus points.

9. Future Enhancements

Multiplayer mode, achievements, daily challenges, friend system, AI opponent, cloud save, mobile app.